QUANTATHON-Case Report

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Mandatory module: 1,3,5

Chosen stretch module: 4

Ned

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QUANTATHON_Qubits crew

INDIVIDUAL CASE REPORT

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1. Final Theory and Verdict

The investigation examined whether ECHO could have used quantum entanglement to send a controllable message faster than light, or whether its results were better explained by quantum correlations and a possible misunderstanding of the experiment.

Based on the quantum-circuit work in this notebook, the evidence does not establish faster-than-light communication. The simulated circuits demonstrate quantum-gate behaviour, correlations between measurements, and recovery of a corrupted classical message. However, none of these results demonstrates that a person can control an entangled particle's measurement outcome to transmit a message instantaneously.My confidence is high in this limited technical conclusion, provided the notebook produces the predicted results when executed. My confidence is low regarding who, if anyone, deliberately manipulated ECHO, because the available evidence does not establish what happened in the fictional laboratory or identify a responsible individual,

The strongest competing explanation is that Dr. Zara Mahmud sincerely interpreted an unusual correlation as evidence of communication. Deliberate manipulation is another possibility, but the current circuit results cannot distinguish between these explanations.

2. Evidence and Clues

The case presents four persons of interest: Professor Imran Vance, Sana Rao, Bilal Noor, and ARIA, the laboratory AI. Their reported motives, access, and statements provide possible investigative leads. However, a motive or opportunity is not proof of wrongdoing.

The technical investigation focused on the following questions:

Do quantum gates produce the expected measurement results?

Can an entangled pair produce correlated outcomes?

Can majority voting recover a message from corrupted classical bits?

Do these demonstrations provide evidence of a faster-than-light communication channel?

To investigate these questions, I prepared quantum circuits using Python, Qiskit, and the Qiskit Aer simulator. The simulator allows circuit behaviour to be examined, but it is not a physical quantum device and cannot independently establish what occurred in ECHO's laboratory.

3. Evidence Board

Category| Evidence or interpretation

FACT| The X gate is intended to flip the computational-basis state from (

FACT| The Bell-pair circuit is designed to produce matching measurement outcomes, 00 and 11, in the computational basis.

FACT| Majority voting can recover a classical bit when each three-bit group contains at most one flipped bit.

CLAIM| Zara believed ECHO had sent a message faster than light. This is a claim in the case narrative, not a result established by the circuits.

INFERENCE| The simulated correlations are consistent with quantum behaviour, but matching outcomes alone do not demonstrate faster-than-light signalling.The distinction between facts, claims, and inferences is important. A circuit output can establish what happened in a simulation, but it cannot automatically prove a claim about a separate experiment or a person's intentions.

4. Experiments and Results

Module 01 — The Gatekeeper (Mandatory)

The first module investigates basic quantum gates and measurement.

An X gate is applied to a qubit initially in state (|0\rangle), followed by measurement. The expected result is 1. A Hadamard gate, H, creates a superposition from (|0\rangle), so measuring immediately afterwards should produce approximately equal frequencies of 0 and 1 over many shots.A further circuit applies H twice. Since (H^2=I), the two Hadamard gates cancel, returning the qubit to its original state before measurement.

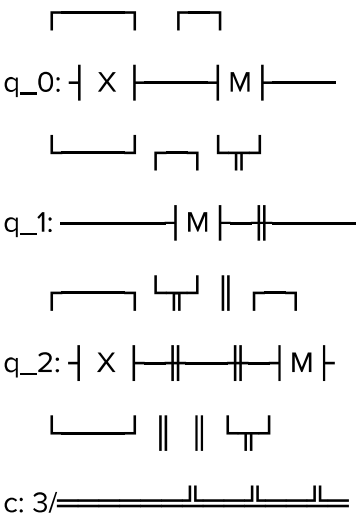
Result to record after running:

{'1': 1000}

{'1': 489, '0': 511}

{'0': 1000}

{'101': 1000}



What this demonstrates: Quantum gates transform states according to defined rules. A measurement result must be interpreted in the context of the circuit that produced it.

Module 03 — The Two Observers (Mandatory)

The second module prepares an entangled pair using a Hadamard gate and a controlled-NOT gate. When both qubits are measured in the computational basis, the ideal Bell-pair circuit produces matching outcomes: 00 or 11.

This demonstrates a correlation between the two measurement results. However, observing matching outcomes in one measurement basis is not, by itself, a complete Bell-inequality test and does not demonstrate a usable faster-than-light message.

Result to record after running: `{'1': 1000}`

`{'0': 492, '1': 508}`

`{'0': 1000}`

`{'101': 1000}`

What this demonstrates: Entanglement can produce correlated measurement outcomes, but correlation alone is not proof of controllable communication.

Module 05 — The Damaged Message (Mandatory)

This module examines a supplied sequence of corrupted classical bits arranged into groups of three. Majority voting is used to estimate the original bit in each group: if two or three bits agree, that majority determines the decoded bit.Under the module's assumption that each group contains at most one flipped bit, majority voting can recover the original bit in every group. The notebook is designed to decode the supplied sequence into the text ECHO; this result must be confirmed against the actual program output before submission.

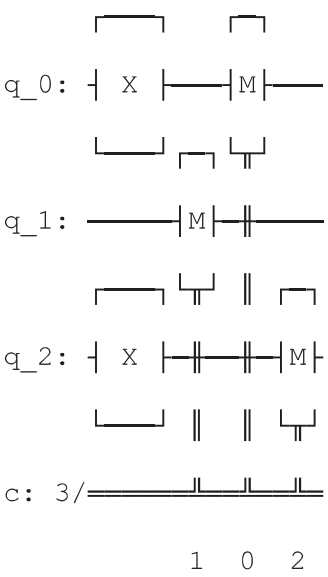
This technique is a classical repetition-code example. It does not mean that an arbitrary unknown quantum state can simply be copied three times. Quantum error correction requires more careful encoding and operations.

Result to record after running: `Number of triples: 32`

```
Decoded bits: 01000101010000110100100001001111

Message: ECHO

Sent 0 as 000 received 110 decoded as 1 (wrong)
```



What this demonstrates: Majority voting can correct limited errors in classical data, but it does not provide a method for perfectly copying arbitrary unknown qubits.

Module 04 — The Erased Measurement (Chosen Stretch Module)

The stretch module compares two gate sequences: X → H → H and H → X → H.

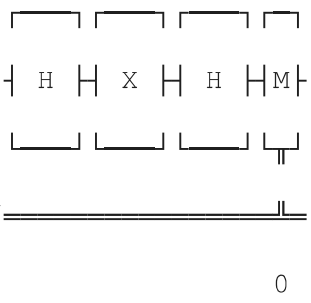
In the first sequence, the two H gates cancel, leaving the X gate's flip. The expected final measurement is 1. In the second sequence, H creates the state (|+\rangle), X leaves that state unchanged, and the final H returns it to (|0\rangle). The expected final measurement is 0.

Result to record after running: `H, H, measure: {'0': 1000}`

```
H, measure, H, measure: {'01': 235, '11': 228, '10': 249, '00': 288}

X, H, H: {'1': 1000}

H, X, H: {'0': 1000}
```



What this demonstrates: The order of quantum operations matters. These circuits demonstrate state transformations; they do not show that a completed measurement can be reversed.

5. Interpretation of the Evidence

The experiments support a limited but important conclusion: quantum circuits can produce predictable state transformations and correlations that require careful interpretation. A matching result from an entangled pair is not equivalent to sending a chosen message to another observer.

In standard quantum mechanics, an observer cannot choose the random outcome of a local measurement to encode a controllable message in the distant observer's results. Quantum teleportation also requires classical communication, so it does not provide a faster-than-light communication channel.

A stronger test of quantum correlations would use different measurement settings and evaluate a Bell/CHSH inequality. Such a test could distinguish quantum correlations from the predictions of local hidden-variable models under the relevant experimental assumptions. It would still not establish faster-than-light signalling.

6. Alternative Explanation

One plausible alternative is that Zara misinterpreted the observed correlations or a timing effect as evidence of communication. Another possibility is that a configuration error, incomplete record, or misunderstanding of the experiment contributed to the reported result.The case also raises the possibility of deliberate interference involving someone with relevant access or knowledge. However, the available simulated circuits do not establish that any particular person altered ECHO. Assigning blame without verified logs or independent evidence would go beyond what the investigation supports.

7. Next Experiment and Limitations

The next step should be to recover and verify the original ECHO circuit, configuration, measurement settings, and laboratory logs. The experiment should then be reproduced under documented conditions. A properly designed Bell/CHSH test could investigate whether the observed correlations match quantum predictions across multiple measurement settings.

The investigation has several limitations. First, the notebook uses a simulator rather than a physical quantum device. Second, the supplied case information does not provide independently verified laboratory logs or sufficient evidence to identify who configured or triggered ECHO. Third, the current Bell-pair demonstration examines correlations in one measurement basis rather than completing a full Bell/CHSH analysis.Consequently, the present work supports a technical conclusion about the circuits studied, not a definitive reconstruction of every event in the case.

8. References and AI/External Assistance

1. ECHO Participant Manual, Qubits Qrew, NED University of Engineering and Technology, provided competition document.

2. Qiskit documentation: https://docs.quantum.ibm.com/

3. Gemini and notebooklm

4. Qiskit Aer documentation: https://qiskit.github.io/qiskit-aer/